Quantum Benchmarks

Hardware Runs

Report issue date: March 23, 2026

This benchmarking report was generated using Classiq v1.6.0.
For further information, see the Benchmarking directory in classiq-library.

adder - 4 qubits

Provider	Backend Name	Score	Time Elapsed (min)
Classiq	simulator	1.0	0.0
Classiq	simulator_statevector	1.0	0.0
IBM Quantum	ibm_kingston	0.204	395.2
IonQ	qpu.forte-1	0.877	15.1

ghz - 3 qubits

Provider	Backend Name	Score	Time Elapsed (min)
Classiq	simulator	0.9972	0.0
Classiq	simulator_statevector	1.0	0.0
IBM Quantum	ibm_kingston	0.4768	379.9
IonQ	qpu.forte-1	0.9609	10.3

grover - 4 qubits

Provider	Backend Name	Score	Time Elapsed (min)
Classiq	simulator	1.0083	0.0
Classiq	simulator_statevector	1.0	0.0
IBM Quantum	ibm_kingston	0.3082	50.9
IonQ	qpu.forte-1	0.6241	10.3

localization - 3 qubits

Provider	Backend Name	Score	Time Elapsed (min)
Classiq	simulator	0.9154	0.0
Classiq	simulator_statevector	0.9147	0.0

qft - 4 qubits

Provider	Backend Name	Score	Time Elapsed (min)
Classiq	simulator	1.0	0.0
Classiq	simulator_statevector	1.0	0.0
IBM Quantum	ibm_kingston	0.6853	86.6
IonQ	qpu.forte-1	0.9973	15.1

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Quantum Volume 2–4 (10 trials)

Provider	Backend	Quantum Volume	Trials	Average Elapsed Time (min)
Classiq	simulator	16	30/30	0.01
Classiq	simulator_statevector	16	30/30	0.01
IBM Quantum	ibm_kingston	8	29/30	439.19
IonQ	qpu.forte-1	8	29/30	12.8

state_preparation - 4 qubits

Provider	Backend Name	Score	Time Elapsed (min)
Classiq	simulator	0.948	0.0
Classiq	simulator_statevector	1.0	0.0
IBM Quantum	ibm_kingston	0.8535	479.3
IonQ	qpu.forte-1	0.8666	10.1

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